

## SORAWEE PORNCHAROENWASE

914-920-1166 ♦ sorawee.pc@gmail.com

### EDUCATION

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| <b>Ph.D. in Computer Science</b><br>University of Washington       | <i>2023</i> |
| <b>B.Sc. in Computer Science</b> (with Honors)<br>Brown University | <i>2018</i> |

### EMPLOYMENT

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| <b>Applied Scientist</b> @ Amazon Web Services, Inc.<br>Applied Scientist II<br>Applied Scientist III | <i>July 2024-Present</i><br><i>Starting July 2024</i><br><i>Starting July 2026</i> |
| <b>Senior R&amp;D Engineer</b> @ Veridise, Inc.                                                       | <i>August 2023-June 2024</i>                                                       |

### EXPERIENCE

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| <b>Applied Scientist Intern</b><br>Amazon Web Services, Inc.            | <i>June–August 2021, June–August 2022</i> |
| <b>Computer Scientist Intern</b><br>Enlearn                             | <i>June–August 2020</i>                   |
| <b>Research Assistant</b><br>University of Washington                   | <i>2018–2023</i>                          |
| <b>Research Assistant</b><br>Brown University                           | <i>2014–2018</i>                          |
| <b>Software Engineering Intern</b> (Release Engineer)<br>Facebook, Inc. | <i>June–August 2016</i>                   |
| <b>Software Engineering Intern</b> (Platform Engineer)<br>Square, Inc.  | <i>June–August 2015</i>                   |

### AWARDS

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| <b>Senior Prize</b><br>Department of Computer Science, Brown University                                               | <i>May 2018</i>       |
| <b>Honorable Mention, CRA Outstanding Undergraduate Researcher Award</b><br>Computing Research Association            | <i>December 2017</i>  |
| <b>The Randy F. Pausch CS Undergraduate Summer Research Award</b><br>Department of Computer Science, Brown University | <i>March 2017</i>     |
| <b>Excellence Award</b><br>VEX Robotics Competition, East Rockaway, NY                                                | <i>February 2014</i>  |
| <b>Bronze Medal</b><br>International Olympiad in Informatics                                                          | <i>September 2012</i> |

### SERVICES

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| <b>Artifact Evaluation Committee</b><br>Object-oriented Programming, Systems, Languages, and Applications (OOPSLA) | <i>2024</i> |
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| <b>Student Research Competition Program Committee</b><br>Programming Language Design and Implementation (PLDI)            | 2024                        |
| <b>Student Research Competition Program Committee</b><br>International Symposium on Software Testing and Analysis (ISSTA) | 2023                        |
| <b>Artifact Evaluation Committee</b><br>International Conference on Computer-Aided Verification (CAV)                     | 2019                        |
| <b>International Collegiate Programming Contest (ICPC) Coach</b><br>University of Washington                              | 2018-2019                   |
| <b>Administrator</b><br>Thai Wikipedia                                                                                    | 2013–2018 ( <i>active</i> ) |

## OPEN-SOURCE CONTRIBUTIONS

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| <b>Racket Code Formatter – Creator</b><br>Developed an extensible Racket code formatter                                                                                                                                               | 2021–Present    |
| <b>Racket – Contributor</b><br>Fixed bugs and improving performance & usability in the core Racket language, the documentation system, the IDE, the website, etc. Administering the server. Moderating the community.                 | 2018–Present    |
| <b>Bootstrap/Pyret – Team Member</b><br>Optimized the Pyret compiler and designing its data science library. Contributed to a block-based editor with a focus on accessibility. Administered the server for the Bootstrap curriculum. | 2014–2018       |
| <b>MediaWiki – Contributor</b><br>Contributed to the MediaWiki software and Pywikibot framework.<br>Developed a static analyzer for an abuse filtering (edit policy) language.                                                        | 2012–2014, 2020 |

## TEACHING

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| <b>Teaching Assistant @ University of Washington</b><br>Computer-aided Reasoning for Software                                                    | Fall 2019, Fall 2021                                                                                                                            |
| <b>Teaching Assistant @ Brown University</b><br>Programming Languages<br>Logic for Systems<br>Accelerated Intro to Programming<br>Linear Algebra | Fall 2017<br>Spring 2016, Spring 2017 ( <i>Head TA</i> ), Spring 2018<br>Fall 2015<br>Winter 2015 ( <i>MOOC</i> ), Fall 2016 ( <i>Head TA</i> ) |

## INVITED TALKS & GUEST LECTURES

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| <b>A Pretty Expressive Printer</b><br>Lean Together 2024                                                                       | 2024 |
| <b>fmt: a Racket Code Formatter</b><br>12th RacketCon                                                                          | 2022 |
| <b>A Formal Foundation for Pretty Printing Expressiveness</b><br>Logic for Systems @ Brown University                          | 2022 |
| <b>An Expressive and Optimal Pretty Printer</b><br>Languages, Systems, and Data Seminar @ University of California, Santa Cruz | 2022 |
| <b>Rosette 4</b><br>RacketFest 2022                                                                                            | 2022 |
| <b>Pain Points in Symbolic Evaluation</b><br>Languages and Translators @ California Polytechnic State University               | 2021 |

## THESES

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- Productivity Tools for Solver-Aided Programming** (Ph.D. Thesis) 2023  
University of Washington – advised by Emina Torlak
- An Inside-Out Resugaring System** (Undergraduate Honors Thesis) 2018  
Brown University – advised by Shriram Krishnamurthi

## PUBLICATIONS

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- Integer Reasoning Modulo Different Constants in SMT**  
Elizaveta Pertseva, Alex Ozdemir, Shankara Pailoor, Alp Bassa, Sorawee Porncharoenwase, Isil Dillig, Clark Barrett  
CAV 2025
- Formally Verified Cloud-Scale Authorization** (Distinguished Paper Award)  
Aleks Chakarov, Jaco Geldenhuys, Matthew Heck, Mike Hicks, Sam Huang, Georges-Axel Jaloyan, Anjali Joshi, Rustan Leino, Mikael Mayer, Sean McLaughlin, Akhilesh Mritunjai, Clément Pit-Claudel, Sorawee Porncharoenwase, Florian Rabe, Marianna Rapoport, Giles Reger, Cody Roux, Neha Rungta, Robin Salkeld, Matthias Schlaipfer, Daniel Schoepe, Johanna Schwartzentruber, Serdar Tasiran, Aaron Tomb, Emina Torlak, John Tristan, Lucas Wagner, Mike Whalen, Remy Willems, Jenny Xiang, Tae Joon Byun, Joshua Cohen, Ruijie Wang, Junyoung Jang, Jakob Rath, Hira Taqdees Syeda, Dominik Wagner, Yongwei Yuan  
ICSE 2025
- FPGA Technology Mapping Using Sketch-Guided Program Synthesis**  
Gus Henry Smith, Benjamin Kushigian, Vishal Canumalla, Andrew Cheung, Steven Lyubomirsky, Sorawee Porncharoenwase, René Just, Gilbert Louis Bernstein, and Zachary Tatlock  
ASPLOS 2024
- A Pretty Expressive Printer**  
Sorawee Porncharoenwase, Justin Pombrio, and Emina Torlak  
OOPSLA 2023
- Rhombus: A New Spin on Macros without All the Parentheses**  
Matthew Flatt, Taylor Allred, Nia Angle, Stephen De Gabrielle, Robert Bruce Finder, Jack Firth, Kiran Gopinathan, Ben Greenman, Siddhartha Kasivajhula, Alex Knauth, Jay McCarthy, Sam Phillips, Sorawee Porncharoenwase, Jens Axel Søgaaard, and Sam Tobin-Hochstadt  
OOPSLA 2023
- Generating Conforming Programs with Xsmith**  
William Gallard Hatch, Pierce Darragh, Sorawee Porncharoenwase, Guy Watson, and Eric Eide  
GPCE 2023
- Testing Dafny (Experience Paper)**  
Ahmed Irfan, Sorawee Porncharoenwase, Zvonimir Rakamarić, Neha Rungta, and Emina Torlak  
ISSTA 2022
- A Formal Foundation for Symbolic Evaluation with Merging**  
Sorawee Porncharoenwase, Luke Nelson, Xi Wang, and Emina Torlak  
POPL 2022
- Fixing Code That Explodes Under Symbolic Evaluation**  
Sorawee Porncharoenwase, James Bornholt, and Emina Torlak  
VMCAI 2020
- Event Loops as First-Class Values: A Case Study in Pedagogic Language Design**  
Joe Gibbs Politz, Benjamin S. Lerner, Sorawee Porncharoenwase, and Shriram Krishnamurthi  
(Programming) 2019

# **CompoSAT: Specification-Guided Coverage for Scenario Finding**

Sorawee Porncharoenwase, Tim Nelson, and Shriram Krishnamurthi

FM 2018